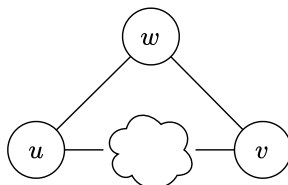


Problem 8.25. An undirected graph is *bipartite* if its nodes may be divided into two sets so that all edges go from a node in one set to a node in the other set. Show that a graph is bipartite if and only if it doesn't contain a cycle that has an odd number of nodes. Let $BIPARTITE = \{\langle G \rangle \mid G \text{ is a bipartite graph}\}$. Show that $BIPARTITE \in NL$.

Part a. Show that an undirected graph is bipartite if and only if it doesn't contain a cycle that has an odd number of nodes.

Proof. For the forward direction, let G be a bipartite graph and show that G doesn't contain a cycle that has an odd number of nodes. The proof is by contradiction. Assume that G contains a cycle $n_1, n_2, \dots, n_{k-1}, n_k$ that has an odd number of nodes. Then, it must be the case that nodes n_1 and n_{k-1} are assigned to different sets because every two consecutive nodes in the sequence $n_1, n_2, \dots, n_{k-1}, n_k$ are adjacent to each other. But this implies that node n_k cannot be assigned to either first or second set as it is adjacent to both nodes n_1 and n_{k-1} , which is a contradiction.

To prove the other direction of the theorem, let G be an undirected graph that doesn't contain a cycle that has an odd number of nodes and show that G is bipartite. The proof is by contradiction. Assume that G is not bipartite. Then, there must exist three nodes u, v and w , such that nodes u and v must be belong to different sets and node w is adjacent to both u and v . This implies that, a path from u to v exists that contains an even number of nodes, and path from u to w is a cycle that has an odd number of nodes, which is a contradiction.



□

Part b. Show that $BIPARTITE \in NL$.

Proof. To show $BIPARTITE \in NL$, we give a nondeterministic log space algorithm M that decides $\overline{BIPARTITE}$. If $\overline{BIPARTITE} \in NL$, then according to Theorem 8.27 $BIPARTITE \in NL$.

$M =$ “On input $\langle G \rangle$, where G is an undirected graph:

1. Let u, v, w, c and m be binary integers stored on work tape.
2. Assign number of nodes in G to m .
3. Non-deterministically assign position of each node in G to u .
4. If degree of u is 0, then *reject*.
5. Let $v = u$, and $c = 0$.
6. Non-deterministically guess each adjacent node of v and store its position in w .
7. Let $c = c + 1$.
8. If $w = u$ and c is even, then *reject*.
9. If $w = u$ and c is odd, then *accept*.
10. If $c = m$, then *reject*.
11. Let $v = w$, and goto step 6.”

□